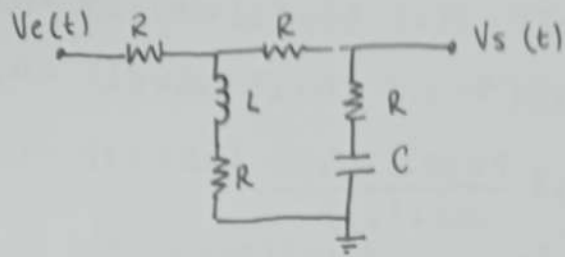


17/09/2026

5.1 Práctica: Diseño de un controlador



Ecuaciones Principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

Procedimiento Algebraico

- Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{(\text{Polinomio } s) I_2(s)}{(?s^2 + ?s + ?) I_2(s)}$$

- 1- Agrupar corrientes $I_1(s)$ e $I_2(s)$
- 2- Expresar el $V_e(s)$ en función de $I_2(s)$
- 3- Despejar $I_1(s)$ de la segunda ecuación principal.

$$V_s(s) = \left(R + \frac{1}{C s}\right) I_2(s) = \left(\frac{C R s + 1}{C s}\right) I_2(s)$$

$$L s I_1(s) = L s I_2(s) + R I_1(s) - R I_2(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$L s I_1(s) + R I_1(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s} + L s I_2(s) + R I_2(s)$$

$$(L s + R) I_1(s) = \left(L s + \frac{1}{C s} + 3R\right) I_2(s)$$

$$I_1(s) = \left(\frac{1}{R + L s} \left(L s + \frac{1}{C s} + 3R\right) I_2(s)\right) \rightarrow \left(\frac{1}{R + L s} \left(L s + \frac{1}{C s} + 3R\right)\right) \frac{C s}{C s}$$

$$I_1(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$V_e(s) = RI_1(s) + LsI_1(s) - LsI_2 + RI_1(s) - RI_2(s)$$

$$V_e(s) = (Ls + 2R)I_1(s) - (Ls + R)I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) - (Ls + R)I_2(s)$$

$$V_e(s) = \left\{ \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1)}{CLs^2 + CRs} \right] - (Ls + R) \right\} I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \left(\frac{3CLR s^2 + 5CR^2 s + 2R + Ls}{CLs^2 + CRs} \right) I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs} I_2(s)}{\frac{3CLR s^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs}}{\frac{3CLR s^2 + (5CR^2 + L)s + 2R}{Cs(Ls + R)}} = \frac{(Ls + R)(CRs + 1)}{3CLR s^2 + (5CR^2 + L)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left(1 - \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \right) = \frac{R}{2R} = \frac{1}{2}$$

$$e(t) = 0.5V$$

Calcular Polos

$$R = 10 \text{ k}\Omega$$

$$L = 220 \text{ }\mu\text{H}$$

$$C = 470 \text{ }\mu\text{F}$$

$$3CLRs^2 + (5CR^2 + Ls) + Ls = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$-235,000.0002 \pm \sqrt{(235,000.0002)^2 - 4(0.003102)(20,000)}$$

$$\frac{-235,000.0002 \pm 234,999.9997}{2(0.003102)}$$

$$\lambda_1 = -0.0851061383$$

$$\lambda_2 = -75,757,575.74$$

$$a = 3CLR = 0.003102$$

$$b = 5CR^2 + L = 235,000.0002$$

$$c = 2R = 20 \text{ k}$$



Transitorio Estacionario

Respuesta Subamortiguada

Modelo de ecuaciones integrodiferenciales

23/09/2026

A partir de las ecuaciones principales del sistema se despejan las variables dependientes, es decir, las corrientes y el voltaje.

$$\dot{I}_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \left] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

$$= \frac{0.001034s^2 + 47,000.00022s + 10,000}{0.003102s^2 + 235,000.0002s + 20,000}$$

$$R = 10 \text{ k}\Omega$$

$$C = 470 \text{ }\mu\text{F}$$

$$L = 220 \text{ }\mu\text{H}$$